

Advances in Explainable AI Since 2019

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Why XAI?

Deep networks are powerful but opaque — and opacity has consequences.

- A credit model that can't explain a rejection may violate consumer law
- A medical diagnostic system that can't be audited may never be adopted
- An autonomous vehicle that can't account for its decisions can't be certified

After 2019: Transformers took over, LLMs emerged, and the EU AI Act made transparency a *legal* requirement.

Post-Hoc Explanation Methods

- **Feature attribution** — SHAP, Integrated Gradients, and others unified under the Shapley framework; TreeSHAP made this practical for tree models
- **Counterfactuals** — “What would need to change?” DiCE generates diverse, actionable alternatives; causal variants respect structural causal models
- **Concepts & prototypes** — ACE auto-discovers visual concepts; ProtoPNet builds prototypes directly into the architecture

Transformers, LLMs, and Mechanistic Interpretability

- **Attention $\neq$ explanation** — raw attention weights are unreliable; attention rollout and gradient-weighted propagation work better
- **LLMs as explainers** — chain-of-thought prompting both explains and improves reasoning, but generated rationales can be *plausible* without being *faithful*
- **Mechanistic interpretability** — reverse-engineers circuits at the weight level; impressive in small models, but scaling to billion-parameter models remains an open problem

Evaluation is arguably the hardest unsolved problem in XAI.

- Faithfulness vs. plausibility: these desiderata frequently conflict
- Application-, human-, and functionally-grounded evaluations often disagree

Emerging challenges:

- Explanations can be adversarially manipulated to conceal discrimination
- Transparency facilitates membership inference attacks — privacy and explainability pull in opposite directions
- Most XAI methods don't scale to foundation models